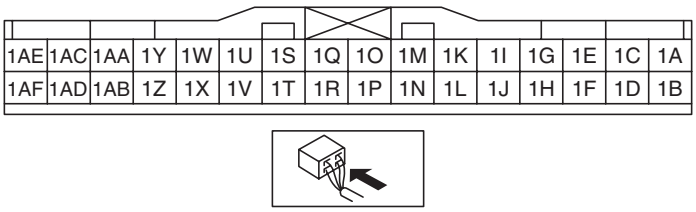
DTC B109E:87 [BCM]

id0902f5347600

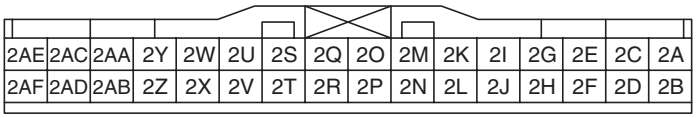
Description	Communication error with start stop unit
Detection condition	• BCM detects a communication error with the start stop unit for **10 times continuously** with the ignition switched ON (engine off or on).
Fail-safe	Not applicable
Possible cause	• Start stop unit connector or terminal malfunction • BCM connector or terminal malfunction • Short to ground in wiring harness between start stop unit terminal 1Q and BCM terminal 2J • Open circuit in wiring harness between start stop unit terminal 1Q and BCM terminal 2J • Start stop unit malfunction • BCM malfunction

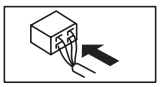


START STOP UNIT
WIRING HARNESS-SIDE CONNECTOR



BCM
WIRING HARNESS-SIDE CONNECTOR





Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY BCM DTCs AGAIN** • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Go to the next step.
		No	Go to Step 8.
2	**INSPECT START STOP UNIT CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the start stop unit connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.

Step	Inspection		Action
3	**INSPECT BCM CONNECTOR CONDITION** • Disconnect the BCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 7.
4	**INSPECT WIRING HARNESS BETWEEN START STOP UNIT AND BCM FOR SHORT TO GROUND** • Verify that the start stop unit and BCM connectors are disconnected. • Inspect for continuity between start stop unit terminal 1Q (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between start stop unit terminal 1Q and BCM terminal 2J. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 7.
		No	Go to the next step.
5	**INSPECT WIRING HARNESS BETWEEN START STOP UNIT AND BCM FOR OPEN CIRCUIT** • Verify that the start stop unit and BCM connectors are disconnected. • Inspect for continuity between start stop unit terminal 1Q (wiring harness-side) and BCM terminal 2J (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between start stop unit terminal 1Q and BCM terminal 2J. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.
6	**VERIFY IF MALFUNCTIONING LOCATION IS START STOP UNIT DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Replace the start stop unit, then go to the next step. (See START STOP UNIT REMOVAL/INSTALLATION.)
		No	Go to Step 8.
7	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
8	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BCM].)
		No	DTC troubleshooting completed.